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Contextual Constraining of Student Design Practices

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Abstract

Background Engineering design is of significant interest to scholarship and engineering educators. As yet, how the higher education context shapes student outcomes in engineering design courses remains underexplored. Since design courses are the primary way students are taught the critical topic of design, it is important to understand how the institutional and organizational contexts shape student outcomes and how we could improve design projects, given the context.

Purpose We sought to answer two questions: What aspects of the design education process are salient, or important, for students? How do these salient aspects affect their design practices?

Design/Method We used a qualitative case study approach to address the research questions because of our emphasis on understanding process-related aspects of design work and developing an interpretive understanding from the students' perspective.

Results Using a nested structuration framework, we show that the context of design practices shaped students' outcomes by constraining their approach to the project and by providing a framework for their design process. We provide recommendations for design educators to help students overcome impediments to achieving learning objectives for design activities. Our research questions the efficacy of teaching engineering design when a design problem lacks a context beyond the classroom.

Conclusions The institutional and organizational contexts influence student design practices. Engineering educators should carefully consider the potential effects of their design projects the implement within a higher education context.

Keywords: engineering design education; nested structuration theory; qualitative case study

Introduction

Design, in its multifarious forms, is an integral component of engineering practices; consequently, engineering education strongly emphasizes the design of systems, components, or processes (ABET, 2011). In spite of the design community's emphasis on design education, teaching and modeling pedagogies remain a challenge because of the inherent complexities of design (Adams, Turns, & Atman, 2003a; Atman, Chimka, Bursic, & Nachtmann, 1999; Conwell,

Catalano, & Beard, 1993; Crismond & Adams, 2012; Dally & Zhang, 1993; Dym, Agogino, Eris, Frey, & Leifer, 2005; Todd, Sorensen, & Magleby, 1993). Studies in engineering education have identified and investigated several local educational contextual factors affecting design. The structure of the project (Moore, Diefes-Dux, & Imbrie, 2005), external support from faculty or clients (Pembridge, 2012), content knowledge, and the team context can influence student engineering design (Lewis et al., 2014; Carberry, Lee, & Ohland, 2010). Although these studies have looked at various factors, they have not attempted to look comprehensively at how the students' context constrains their design practices and what, within their context, are the salient, or important, factors that shape their designs. The complexities of design are not limited to the constructs of design practices, but are situated within larger contexts (Cappelli & Sherer, 1991). Describing various contextual factors helps to explain the constraints or opportunities for the situation under investigation (Johns, 2001; Rosseau & Fried, 2001). Understanding how the context shapes students' design practice, specifically through constraining it, is important since students are never fully removed from the effects of the classroom or institution.

In this article, we focus on the role of constraints in the design process, but take a novel view by incorporating the constraints into the environment where design takes place. We examine how the context in which students undertake their projects – their course, school, physical space, and available resources – shapes their practices, decisions, and outcomes. Although empirical studies of design processes are common (for a review, see Mehalik & Schunn, 2006), studies on the effect of context on design practice in an educational setting are limited. Results from investigating the contexts of the educational setting demonstrate the effect of these contexts on student practice within the context of an engineering course. To understand the effects of the educational setting, we studied engineering design teams within the context of their engineering design project. Two research questions guided our investigation of how students undertake design in an educational context: What aspects of the design education process are salient to students? How do these salient aspects affect student design practices?

Overall, this qualitative study aimed to understand situated student design practices within a specific engineering education context to illuminate and demonstrate the role of contexts and constraints on design. We first review related work and describe the theoretical framework that guided our research study – both its background and how it guided our analysis of the data. We then present the findings on the various aspects of the design project and the types of constraints that affected student practices. We discuss the findings with respect to how they can inform design curricula and frame design projects within an educational setting. We also discuss the practical and theoretical implications of this work. Finally, we provide recommendations for design educators to incorporate into design projects that can aid students in achieving the desired learning objectives.

Literature Review

The context of design is complex, situated in real-world scenarios, with certain degrees of freedom (Jonassen, 2000). One mechanism for reducing this complexity is to apply constraints to generate a pathway towards achieving successful design outcomes (Wulf, 1998). Within a conceptual design space, where designers think about possible solutions, constraints function primarily by limiting and thus directing and structuring design choices (Newell & Simon, 1972; Stokes, 2001). While constraints provide some guidance (Chua & Iyengar, 2008), designers still face numerous choices. When students are faced with too many choices, their evaluation and selection processes become costly in terms of the resources used; therefore, optimal constraints

are critical for a successful design outcome (March & Simon, 1958; Ornaheim & Biskjaer, in press ; Schwartz, 2004).

The role of constraints is important to the study of engineering design because designers face inherent limits within their search of possible design solutions. Constraints, which are clearly defined limits, assist in the design-making and selection process (Dym & Little, 2009). Constraints can emerge, or can be derived, from the context of the problem or task (Daly, Adams, & Bodner, 2012). In well-structured design problems, constraints are well defined and restrict solutions. Categorizing a design problem, for example, as a heat transfer problem, further constrains the possible solutions and methods used to arrive at the final solution (Jonassen, 2011). Constraints associated with classifiable concepts are used extensively in formal design education (Jonassen, Strobel, & Lee, 2006).

Just as designers utilize constraints in their design practices, the environment or context where design takes place is constrained. Bucciarelli's (1994) ethnographic study of engineers illuminated engineering design as a situated and social process rather than an isolated and purely factually based process. The situated and social nature of design is relevant to the investigation of engineering students' design practices because the environment where students practice design is also situated and social (Downey & Lucena, 2003). For instance, in a mechanical engineering course that structured the lectures and assignments around problems rather than topics, Jonassen, Khanna, and Winholtz (2011) found that students perceived the course to lack structure and had difficulty adjusting. The methods students used to study for exams were based on their expectations of an educational setting, and students had difficulty adjusting to the different methods needed to solve the problems. The educational setting is structured in such a way that students' work must be evaluated on some criteria, and the assessments of student work are typically easier when they are well defined. Students and instructors become accustomed to the methods used to complete and evaluate well-defined assessments because of the assessment criteria's fit within the educational structure.

The constraints of an educational setting also impose various compromises that need to be considered when designing assignments or activities. For example, activities that promote enhanced student learning and engagement often require more time and material resources than lecturing (or a traditional classroom structure) from the instructor or program. Instructors must consider the relationship between achieving student learning outcomes and the associated resources when designing assignments and activities. Williams, McNair, Crede, Paretti, and Terpenny (2010) developed a classroom activity that aimed to minimize the tradeoffs between achieving engagement and costly resources by establishing the context for the activity. Williams et al. identified the related learning objectives and constraints of the educational setting – similar to how engineers solve engineering design problems – and what objectives could be met given the constraints. In the Williams et al. (2010) study, the constraints of the educational setting played a role in structuring classroom activities that met certain objectives.

Dannels (2000) and Paretti (2008) highlighted the prevailing influence of constraints associated with the academic context on student design practice, specifically communicative practices in design. While the need to achieve good grades motivated many of the student design practices (rather than the real, or professional, context of designing for clients), students still learned design through the academic context. In other words, students worked within the constraints and adapted to the context of the learning situation.

Dannels (2000) suggested students' adaptability to work within the constraints of an academic context will transfer to the professional context when these individuals are part of the

workplace, or perceive the professional context as real. Paretti (2008) indicated that instructors can apply the academic context in helping students to develop communication skills by being explicit about the needs, goals, and constraints associated with communicative design practices in specific contexts. Instructors can also apply the structure of assignments within the academic context to support student practice. By using constraints to sequence assignments associated with a design project, instructors can support students to stay on track and develop design artifacts over the progress of the project (Paretti & Burgoyne, 2005).

When a course is designed to include a considerable amount of reflection and attention to learning, the design of the course does not always guarantee that student practice will follow this model. Newstetter (1998) observed and studied a student team in a mechanical engineering design class where the pedagogical views of the instructor and the classroom views of the students were often in conflict; this conflict resulted in the students' failure to learn design. That is, the students' tendency to view design activities as merely academic exercises led to a divide-and-conquer approach to team activities, and students were only motivated to get a good grade (Newstetter, 1998). The students' view impeded their utilization of design process components and tools that were intended to scaffold their learning and completing the design activities. Students viewed the design activities as academic exercises or busy work, and the instructor's "learning view" (p. 122), or pedagogical intentions, was lost on the students, whose attention was on performance and grades rather than understanding.

Research Framework: Nested Structuration

Several studies on engineering design in educational settings utilized theoretical frameworks that highlighted how meaning is socially constructed in specific contexts (Dannels, 2000; Paretti, 2008). In this study, we apply the theoretical framework of nested structuration (Perlow, Gittell, & Katz, 2004) because it integrates both the inhibiting yet enabling feature of constraints and outlines ways that individuals interact as part of their context or social system. Perlow et al. (2004) studied work patterns of software developers by examining workgroup interactions and how the larger societal contexts (the organization and institution) affected these patterns. Figure 1 illustrates the nested concept and the mutually reinforcing relationships at each level. Perlow et al.'s findings suggested that not only did the organizational context affect patterns of interaction between group members, but also the organizational system was further reinforced by the overarching institutional context.

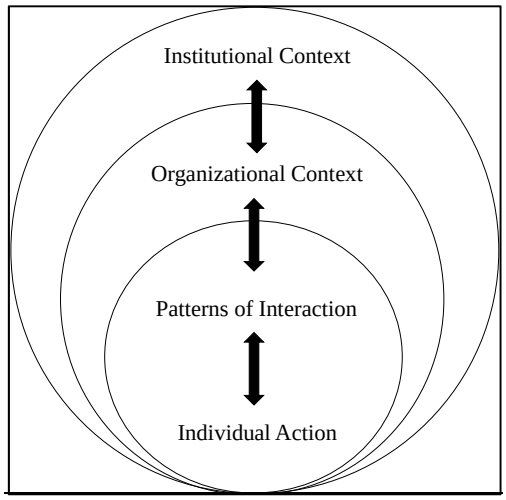


Figure 1. Nested structuration (Reproduced from Perlow, Gittell, & Katz, 2004. p. 533)

Studies that have examined design and engineering workplace contexts have appropriated the nested structuration model to explain the effect of environment on practice; however, no studies incorporated the educational context. Hinds and Lyon's (2011) study on how design thinking and practice are embedded in the cultural context revealed that factors at the institutional level, such as client expectations, affected thinking and practice at the individual level and the design thinking and practice varied across cultures. The teaching-learning ecologies explained in Bailey and Barley's (2011) study of engineering firms treated learning as the missing link between the environment and structure, and demonstrated how environmental constraints shaped organizational structures through actions and practices of the individuals working within the environment.

We used Perlow et al.'s (2004) nested structuration model to explain the relationships between the individual, the organizational context, and the institutional context in an educational setting because the original model was derived from an attempt to contextualize findings regarding workgroup interaction. In this research, we recognized the importance of context for students working on engineering design projects in an educational setting and posited that the nested structuration theoretical framework would help to explain how the organizational and institutional contexts associated with an educational setting affect design practice. Contextualizing the findings of our study under the nested structuration model illustrated how the factors present in multiple levels of context affect student design practice. Nested structuration allowed us to consider how practices are shaped by the larger contexts and illustrated the value of analyzing the factors associated with each context. For instance, if we were to study only design practice as part of a design process model, we would fail to uncover any effects of the course or university. The treatment of design practice as a distinct organizational or institutional component ties together design practice and the contexts associated with the design project, the engineering course, the institution, and the interconnections that exist between them. In addition to using the nested structuration framework to frame our analysis, we followed Perlow et al.'s approach of studying engineering work groups in their organizational and institutional contexts to investigate factors that affected student design practice at multiple levels.

Research Design

Qualitative Case Study

Research methods including protocol analysis, interviews, surveys, ethnographies, and analysis of learning artifacts are used in engineering design education research (Atman, Eris, McDonnell, Cardella, & Borgford-Parnell, 2014). Ethnographies, typically exploring practices or experiences in situ, are a naturalistic approach for studying engineering students (Tonso, 2006) and how students engaged in practices such as design (Downey & Lucena, 2003; Garzca, Palou, Lopez-Malo, & Garibay, 2009). Case studies satisfy the three principles of qualitative research – describing, understanding, and explaining – and are used to examine the how and why aspects of an investigation (Yin, 1994).

We chose a qualitative case study approach, where our unit of analysis was a student design team. Employing an interpretive epistemological perspective (Van Note Chism, Douglas, & Hilson, 2008) was useful because we intended to contextualize aspects of the design education process that are salient to students. To establish trustworthiness in our study, we provided thick descriptions of the research setting and context, triangulated our data, used multiple sources, and used separate coding as part of the data analysis (Creswell, 1998; Leydens, Moskal, & Pavelich, 2004). The following sections describe the study's participants, course instructor, engineering course, and the design problem in order to provide a detailed, context-specific description of the research setting. We explain how we accounted for rival conclusions and alternative explanations in the Discussion section of this article.

A benefit of our approach is the potential for a better theoretical and empirical understanding of the interrelated activities within a case that is offered by the qualitative case study's holistic approach to investigating phenomena (Feagin, Orum, & Sjoberg, 1991). As another mechanism to ensure trustworthiness, we studied the design teams using multiple sources of evidence that included video observations of team design meetings, student work, and interviews (Feagin et al., 1991; Merriam, 1988; Miles & Huberman, 1994; Yin, 1994; Stake, 1995). In addition to the data or evidence collected from the design teams, we collected course artifacts, such as the design project documents. For an additional source of evidence, we interviewed the design teams at the completion of the project. A more comprehensive perspective on student design practices is fundamentally important to understanding the salient aspects of the design project and to answer the research questions of this study.

Participants and Research Setting

Student participants All individual participants were first-year students who were enrolled in the first-semester engineering course. This course consisted of a separate lecture and semester-long supplement course to the main lecture. Approximately half of the students (16) enrolled in the supplement course chose to participate in our study. In total, participants included one female and 15 male students. The gender composition for the supplement course reported in this study was 14% female, 86% male. To form the design teams, the instructor randomly placed the students who volunteered into teams and students who did not volunteer into different teams. In this study, each team consisted of four students. Participant selection was guided by an attempt to study the experiences of student design teams in a natural setting through the observation of design team sessions, so the selected teams were formed the same way as the teams who did not volunteer to participate.

Engineering course The design project we studied was a main assignment of the engineering course, where students worked on the project eight of the 15 weeks of the course.

Students typically take this engineering course in their first semester as part of the required engineering program curriculum. The objectives of the course include problem solving, design, modeling and visual representations, algorithm development, programming, ethics, and professionalism. The instructor guided teams with class material and discussions, and met individually with teams. The instructor graded the 16 design assignments. A rubric accompanied each design assignment, and the students knew the point allocation and who graded the assignments.

In addition to the engineering course discussed in this study, students were also required to take other math and science courses as part of the engineering curriculum; they typically had a math-, science-, and engineering-heavy course load (11 of approximately 17 credits, according to the university's suggested schedule for engineering majors).

Course instructor Instructors for the course were typically graduate engineering students from various disciplines and were responsible for instruction and assessment of the course material and design project. All instructors received training from experienced faculty members in engineering education before the start of the semester and participated in weekly group sessions regarding the course material and administration. The teams participating in the study had the same instructor and received the same material and instruction as part of the course. The instructor for the students who participated in this study was a master's-level graduate teaching assistant. The instructor's background was in civil engineering and water resource engineering. Her one and a half years of experience teaching the course included design team mentoring, assessment of design projects, and instruction on engineering design.

Design Problem

The nature of the design project was ill-structured (lacking definition in some aspects), but the goals and constraints associated with structure of the project were more explicit. A design document accompanying the project described the objectives and requirements for each design assignment, for example, the prototype. For each of the assignments, the document also included templates, which provided guidelines for what to include and formatting. The prompt for the project was generally open-ended: develop a design solution that creates awareness of a renewable energy source. A subsection of the design document delineated more specific parameters of the project for students. Table 1 lists the parameters given in the project design document, other parameters considered by the design teams not given in the document, and definitions for all parameters. The $20 budget for their final design included a penalty, or percentage deducted from the total score in the rubric, for going over-budget. In our analysis, we emphasized the given parameters because it was important to understand how students approached the design process and how the given parameters affected subsequent practices. The role of these parameters, as they related to student design practice, was incorporated in the data analysis and is discussed later with respect to nested structuration.

Table 1 Design Project Parameters

Parameter	Definition or Explanation
Functional	Design should showcase the creation of energy from a renewable source, not just as a conceptual model, but as a physical implementation. Even though you may not be able to make your design full-scale, your project should adequately demonstrate the intentions of your group's design when scaled up.
Safe	Your design should not harm anyone who uses it, and should have a minimal adverse effect on the environment.
Innovative/interesting	Be creative! Think outside the box on this project. Using a solar panel that you get in a science project kit to power a light bulb is not very interesting. Make your design unique.
Renewable energy source	Design must highlight one or more key components of a renewable energy source.
Educate/entertain	Design should strive to educate and entertain as well as generate further inquiry and interest in renewable energy sources.
Possible/applicable	A possible and applicable solution that benefits or addresses the needs of a user.

Data Collection

We collected the data from a variety of sources including observations, interviews, and the document associated with the design assignments; we had approval of the university Institutional Review Board and complied with their regulations. We had no direct involvement with the course or student participants outside of the study. We video recorded student design activity and took observational field notes during team design sessions in a predetermined meeting room provided by the research team, which included the authors and undergraduate research assistants. Later, the research team transcribed the video recordings verbatim and edited the transcripts to include notes of when participants were doing tasks that did not involve a normal amount of speaking among the participants; such tasks included prototype building, sketching, and using various computer programs. In total, across the four design teams, we transcribed and recorded 29 hours of video observations. Twenty-one design meetings outside of class, consisting of eight categories of meetings (research report, brainstorming, post-brainstorming, prototype discussion, prototype building, final design report, final design building, and final presentation preparation), and four in-class presentations contributed to the dataset.

At the completion of the project, we interviewed three of the four participating teams; we aimed to gain an insight into the student perspective regarding their experience with the design project and course. We were not able to collect interview data for Team 3 because participation in the interview was on a volunteer basis. The interviews were semistructured and used open-ended and follow-up questions (see Appendix for the protocol). The open-ended and follow-up questions allowed interviewers to elicit more elaborated responses. Teams initially responded

that the budget was the main parameter considered when making decisions for their final design. From the follow-up responses, we were able to identify other factors, such as creativity and peer opinion, that were part of some teams' decision-making processes for creating their final designs.

Data Analysis

We focused on explanation building (Yin, 2009) and utilized within-case and cross-case analysis (Miles & Huberman, 1994). Nested structuration guided our approach to examining the video observations and transcripts. Eisenhart (1989) suggested a two-part analysis where in the first step the researcher analyzes the within-case data by listing events and critical incidents and/or creating taxonomies and networks of the data, and in the second step, looks for explanation and causality. We used Eisenhart's analysis by applying the coding framework, which was a result of a two-part approach. Two coders from our research team separately analyzed the video data and applied the appropriate codes to each segment. We employed multiple coders to help increase the trustworthiness of the outcomes of this study.

The coding framework evolved to include three principal codes and several subcodes derived from each of the principal codes. We developed the three principal codes according to the constructs of the theory of nested structuration. The principal codes focused on design, technology, and the design project. This study specifically focused on the design project code for the analysis to help answer the study's research questions. The design project code had seven subcodes, and we applied it to ideas related to the design project as well as the organizational and institutional contexts. The main constructs derived from nested structuration were broken down into several subconstructs related to the data. Table 2 lists the constructs, subconstructs, and instances of these constructs.

Table 2 Coding Framework

Construct	Subconstructs		Instances
Institutional			Curriculum Course load/workload Infrastructure
Organizational	Course (Level 1)		Course tests, quizzes, assignments Grades Homework Instructor
	Project (Level 2)	Structure (Level 2a)	Design project requirements Design project objectives Grading Cost Timing/design assignment deadlines
		Parameter (Level 2b)	Functional parameter Safety parameter Interesting/innovative parameter Renewable energy source parameter Possible/applicable parameter

Individual	Interaction-patterns	Decisions Discussions Divisions of labor

In order to understand the dynamics of each team, or case, we analyzed the observation and interview data from individual teams and across teams. We assigned pseudonyms to the student participants, and all reported quotations are identified by a pseudonym. On the basis of results from individual cases, we identified common themes or patterns that existed across design teams in this study. Guided by the constructs of nested structuration, we later discuss how these common themes affected student practice at the institutional, organizational, and individual levels.

Findings

We present findings from the within-case and cross-case analyses. The results illustrate how organizational and institutional contexts affected student design practice. According to the nested structuration framework, aspects that were salient to students existed at the institutional level, and at three levels of the organizational (course, project structure, and project) context for each team in this study. We summarize the individual case analysis here due to space limitations. We highlight a few salient issues and follow up with a detailed cross-case analysis. For a full narrative of the design cases and the set of analyzed data, see Goncher (2012).

Within-Case Findings

Table 3 presents a compilation of the salient factors for design teams based on the institutional and organizational contexts. Several factors were common across teams for each of the different contexts, and other factors were not salient for every team.

Table 3 Salient Aspects of the Institutional Context and Organizational Context

	Team			
	1	2	3	4
Institutional context				
Workload	✓	✓	✓	✓
Institutional infrastructure	✓	✓	✓	✓
Organizational context (Course, Level 1)				
Student perception of the course	✓			✓
Organizational context (Project structure, Level 2a)				
Project deadlines	✓	✓		✓
Requirements and objectives	✓			✓
Organizational context (Project parameter, Level 2b)				
Functional	✓	✓	✓	✓
Possible/ applicable	✓	✓	✓	✓
Interesting/ innovative				✓

Institutional context The findings from the within-case analysis demonstrated that teams are never fully removed from the context(s) in which they design. With respect to the institutional context, teams continually discussed coursework not related to the design project. Students' attention to work unrelated to the project affected their approach to completing design assignments such that students completed the assignments perfunctorily to meet the requirements in order to get back to studying or completing other assignments. Students had frequent and comparative discussions regarding the number of credits, what courses, and what work they had due for these other courses within the institutional context. Teams in this study discussed the previous design project experiences of other students, who we identified as part of the institutional context. Teams often used the information obtained from other students to inform their design decisions or approach to the project. Findings from our analysis of the institutional context are explained in more detail as they related to all teams in this study in the Cross-Case Findings section below.

Organizational context The findings showed that the organizational context existed at the course level (Level 1) and project levels (Project structure, Level 2a and Project parameter, 2b) and that students based their design approach on their perception of the engineering course. Some teams perceived the course as a weed-out course and subsequently treated the design project as another academic exercise. A weed-out course is an introductory course that students typically describe as difficult and believe is designed to ensure they are serious and willing to apply the time and effort needed to pass. Students' treatment of design project activities as merely an academic exercise is contradictory to the desired outcomes of a design project, which include applying and learning about the design process. Educators typically implement design activities in their courses with the intent of creating meaningful activities that interest students. Students' perception of a weed-out course shaped some students' view about the learning aspect of the course described in this study.

Some students viewed the course as an exercise in learning to pass, rather than viewing the assignments as part of learning the design process. Teams who viewed the course as "all about learning how to pass" (Jason, Team 1) tried to interpret and predict instructor evaluations. Teams subsequently completed the design assignments on the basis of on their perceptions of the instructor and her evaluation of the assignments.

At the project levels (Project structure, Level 2a and Project parameter, 2b), the structure and parameters of the project affected student design decisions. Overall, when the students were uncertain about the project objectives and requirements, most teams chose to focus on the explicit constraints of the design project and often circumvented or evaded the design assignment directions by making up the reported results. Table 4 summarizes salient aspects for each team and resulting practices related to the constructs of nested structuration.

Table 4 Resulting Student Practices Associated with the Levels of Nested Structuration

	Team			
Context	1	2	3	4
Institutional	Coursework associated with other courses and a roommate's previous experiences influenced design project decisions.	Used a regular meeting schedule to improve time-management issues. Previous students' experience with the course to informed their approach to the project.	Perfunctorily created design assignments to get back to other coursework. Previous students' experience with the course informed their approach to the project.	
Organizational (Course, Level 1)	Perfunctorily created design assignments to complete the assignments associated with the project.		Relied on assumptions of the instructor and design assignment evaluation to direct design practices.	Perceptions of the instructor and the course as a weed-out course influenced design practices.
Organizational (Project structure, Level 2a)		Used design assignment requirements to guide their subsequent design practices.	Used design assignment objectives and requirements to guide subsequent design practices.	Followed step-by-step process of the project that led to fabrication of some results.
Organizational (Project parameter, Level 2b)	Focused on design solutions that were functional and possible, but adjusted the cost constraint.	Focused on making a functional device that was applicable and beneficial on a larger scale.		Focused on design solutions that were interesting and beneficial on a larger scale.

Cross-Case Findings

The cross-case analysis of teams identified the common salient aspects across the teams and described how those aspects affected student design practice. These findings demonstrate that influential aspects occurred at the institutional and organizational levels.

Institutional context: *Workload* Students' course load (and resulting workload) was salient across all teams and was present in design session discussion, even at times dominating the discussion. The course load also affected their design practices related to the design project. Teams 1, 3, and 4 discussed courses not related to the engineering course and/or the design

project (this was also true for Team 2, but it was comparatively minimal). These teams compared the amount of coursework as well as the number of credit hours, and the discussions would typically develop to include additional comparisons of the instructors and assignments or tests. Students used comparisons to gauge their own progress and advise one another on the amount of time and effort required by assignments.

The data also showed that not only did these recurring discussions arise while students worked on their projects in design sessions, the discussions also affected how they completed and assigned work. Teams used the designated design session times to complete the assignments required by the design document and decided how to divide tasks on the basis of the institutional factors. Team 1 assigned sections of the research report on the basis of the perceived amount of effort required to write each section coupled with the amount of work outside of the design project each member had. The division of labor based on workload was one approach to completing design assignments. As Eddie explained, “Cause I would just say, if you're not that busy this week, then you would take the longer one. That makes more sense” (Team 1, Research Report Session).

Other teams used the amount of other coursework to guide their approach to writing the research report, but used the shared time together during design sessions to work on and complete design assignments with the assistance of one another. Teams generally used this collaborative approach where one teammate would help another if both students finished their own task before the assignment was completed. The collaborative approach was applied to make the design sessions go faster so students could return to studying for other courses or completing other coursework. As Brett stated, “I'm trying to get this done so I can go back to studying for Physics” (Team 3, Final Design Report Session).

The effect of coursework led other teams to not submit or completely forget to do additional design assignments. For example, a member from Team 2 did not complete a required team evaluation because he was preoccupied with another course project that was due at the same time as the team evaluation. As Gary explained, “Like, I had some computer science project due that night right after the test so I wasn't even thinking about it” (Team 2, Interview). In one instance, students’ focus on other coursework led one student to plagiarize his section of the project research report. As Eddie explained, “No, it's completely my fault. My bad. I just had 17 things to do, so I was just like, maybe I can get away with this. Clearly I didn't though” (Team 1, Brainstorming Session).

Institutional infrastructure In addition to the amount of coursework from other courses within the institutional context, the structure of the institution affected student design practices. Design teams in this study often used information obtained from roommates, hall mates, or other students. Teams 1 and 3 used information about other students’ prior design projects and experiences to inform their selection of the renewable energy source, and how to approach the graded design assignments. Team 1 misrepresented their data and actions in order to fit the project requirements. For example, Cory recounted his roommate’s experience of misrepresenting the team’s actions: “My roommate said too, for stuff they – they bought stuff and theirs ended up being $21, or something. They just went on eBay and found the price and printed out a receipt” (Team 1, Prototype Session). Team 3 used information from a student who took the course the previous semester to interpret how they approached the team evaluations. When asked how they should evaluate one another according to their contributions at the time of the evaluation, Ian referred to what he learned from another student: “Nah, you're straight.

Someone from last semester, he told me, everyone just gives each other hundreds [percent]. Otherwise it'd be [shakes head side to side]" (Team 3, Research Report Session).

Team 4 illustrated the peer-informed approach when writing the research report. Here, they utilized information from another student to guide how much effort they were willing to put into one design assignment. The design team discussed if the instructor would be too strict on the paper; they mentioned that since the instructor has to read so many papers, her critiques on individual papers would not be that harsh. As Craig explained, "Yeah, [student] said don't even worry about this thing, just throw it together and go with it. He lives right next door to me in the dorm. But he's a second year and he did this" (Team 3, Research Report). Team 2 did not seek out information from students within the institutional context to interpret project requirements; however, they referred to other known students' experiences in order to try to clarify where they could acquire resources or materials. When discussing how they would obtain a motor for their project, Walt hoped they would find out from the instructor. Walt referred to his roommate's experience from the previous semester, but did not directly ask his roommate for assistance.

An institutional factor, such as curriculum structure, is typically intended to influence student practice by providing students a structure that incorporates fundamental courses and technical knowledge. Data from teams' design sessions showed that students in this study took physics and math courses. Students did not specifically incorporate physics and math content knowledge into their design practices; however, the design project did not require incorporating specific content knowledge. Student perceived the engineering-related knowledge from these courses as insufficient for their design approach. Team 4 felt that after taking other engineering courses after their first-year, they would be able to produce a more functional and creative design. Team 3 rejected any design solution they thought was too technical and perceived that they did not have enough engineering-related knowledge to produce the final design. The findings related to curriculum structure suggest the compartmentalized nature of the institutional context constrained students from fully applying material they did not view as technical or engineering-specific knowledge. At the same time, the nature of the design problem may have also contributed to students' approach because it did not specifically require the design teams to apply content from their mathematics and science courses.

Organizational context (Course, Level 1) *Student perception of the course*

Students' perceptions of the course influenced their design practices. Teams 1 and 4 expressed their view of the course regularly during their design sessions and interviews. Characteristics that students attributed to weed-out courses included instructors making the course "ridiculously hard," requiring excessive amounts of work (usually tedious work), and making it difficult to pass. Teams 1 and 4 perceived the course as "a bunch of tedious stuff" (Craig, Team 4) or "all about learning how to pass" (Jason, Team 1). The weighting of the course (in comparison to their other courses) factored into how students view of the amount of required work for engineering course. As Channing explained, "I'm in multivariable Calculus and Physics [course number], so I really have to study for those classes, and I was spending all of my time on this two-credit weed-out class" (Team 4, Interview). Other courses in an engineering student's suggested schedule were typically three or four credits.

A weed-out perspective of the course affected how students approached the design project in terms of how they thought about the project and completed design assignments. Because they treated the design assignments as academic exercises, students completed the assignment with marginal exploration of the design phase. Channing separated his view of the

course and his practice within the context of the course from how he would react in a professional or real-life situation. Channing stated, “We're not doing that bad. If it comes to a real project, I'd talk to the company and be like, ‘Yeah, this isn't safe’, but this is a [course] weed-out class design project” (Team 4, Final Design Presentation).

The data from Team 3 suggested that this team did not perceive the course as a weed-out course, but more of a typical [engineering] course. In other words, Team 3 did not refer to any of the characteristics associated with a weed-out course (“ridiculously hard”) that required abundant work to discourage students. Team 3 consciously applied or utilized certain course material on design, such as idea triggering, during the brainstorming design phase. Team 3’s perception of a typical course led them to revert to practices typical of a student participating in any type of course, usually with the primary goal of getting a good grade on an assignment and in the course. During the creation of design assignments, students would use their perception of the instructor to guide how they approached the design assignments. For example, Team 3 chose to include more definitions and specifically wrote about how they utilized the idea-triggering method in order to elicit positive instructor feedback and stated that the instructor would “eat that up.” Team 4, on the other hand, discussed their perceptions of the instructor and decided what they needed to do on the basis of their evaluations of the instructor. In this case, they decided not to check for grammar on the research report. Cody rationalized that the instructor was “obviously not an English major” (Team 4, Research Report).

Organizational context (Project structure, Level 2a) *Step-by-step nature of the design project*
The step-by-step deadlines for the design project affected the approach of Teams 1, 2, and 4 to design. The sequence of the due dates and associated design assignments matched the design process given to students in class. Students viewed the step-by-step nature of the project as more of an approach that facilitated time management rather than an approach that encouraged them to concentrate on each phase of the design process. As Walt explained, “And the way we had certain stuff due at a certain time worked well. It made it, like, so it didn’t stack up on us, so it had to progress with the project. So I liked that” (Team 2, Interview). The deadlines for each design assignment enabled teams to fit the design project in with their other coursework. Students viewed the deadlines as a benefit in the organizational context. Having each design assignment due sequentially – in weekly or biweekly intervals – enabled them to manage their time so that the project “didn’t stack up” (Walt, Team 2, Interview) on them and “made it less stressful” (Eddie, Team 1, Interview). The linear deadlines for assignments in this study did not encourage students to revisit any of the design phases or approach design as an iterative practice.

Highly constrained deadlines even led students to misrepresent their actions related to explicit constraints, such as the project deadlines. Team 4 described how the due dates for the final design report and final presentation prevented them from reporting an accurate account of how they constructed and tested their design; they stated that the final design report required a picture of their design and they hadn’t even finished or tested their design at the time of the report due date. Furthermore, they felt that they were forced to misrepresent pictures of the final design and bill of materials because they were “going to be graded on it” (Channing, Team 4, Final Design Report Session). By fabricating their reported actions to meet the requirements for the final design report, in order to submit a completed report by the required deadline, the team chose to fulfill the final report requirements without fully understanding the relationship among the associated phases of the design process.

The linear progression of design phases, which matched project deadlines, did not encourage – and actually constrained teams from – revisiting design phases or making connections between these phases. In an example from Team 2, some team members did not make the connection between writing the research report and how it related to the "gather information" phase of the design process that they were required to write about in the final design report. They also skipped the "define the problem" phase of the design process early on because they were not explicitly required to do so in the design assignments that were due earlier: "So I guess, in that sense, we didn't really define a problem" (Jim, Team 2, Interview).

Design assignment requirements and objectives One project requirement was the bill of materials to prove the project cost under $20. The explicit budget constraint and the requirement to report on it led all teams to focus on how they would keep the cost under $20 and then to falsely report that their design was under-budget in the final report. The following excerpt from Team 4's discussions illustrates the finding that teams did not truthfully report their actions. Channing stated,

> We're not putting down, "matching printed circuit board." We're just saying "breadboard" and then we're going to reference Radio Shack. I don't think she's going to be checking all of our shit, so. This is cost adjustment. (Final Design Report)

Many of the teams exhibited a sunk-cost perspective in terms of time and money due to the explicit constraints of the design project. Teams would buy additional materials and supplies if their design concept did not work, instead of going back to earlier phases to redesign. They would then report what they could adjust to fit within the budget. Team 4 purchased a high-speed motor and realized they needed a low-speed motor after testing. To adhere to the budget, they reported buying and using the high-speed motor with no documentation of the actual component used in their final working model.

Another outcome of the design assignments was that the objectives and requirements enabled teams, at times, to relate the design phases to one another. For example, the sketching assignment required students to provide a detailed solution and short explanation. The initial response of teams was to make up something for the required explanation. However, the teams' attempt to make s up something prompted a discussion where teams thought about how they arrived at the design solution. In Team 3's sketching session, they initially suggested making up the explanation; but after thinking through what to include in their made-up description, they concluded they chose the final design idea because their idea generation led them to consider small household appliances. The more truthful and reflective outcome was also found in cases when teams used the sketch to discuss and hash out features of their conceptualized design to inform building in later phases.

Organizational context (Project parameter, Level 2b) This section discusses the common themes related to the design project parameters (see Table 1), as defined by the design document, and other emergent parameters that the teams considered.

Utilization of a renewable energy source While all teams utilized a renewable energy source in their design solution, incorporating the renewable energy source was not the main focus when trying to conceptualize design solutions. Because teams used a common list of possible renewable energy sources, they produced similar ideas during the brainstorming

sessions. For example, Teams 1 and 3 both contemplated an idea that used a renewable energy source to power road lights. The teams' focus was on the benefits of road lights, not the utilization of the renewable energy source. Team 1 considered using geothermal as a source, and Team 3 considered using either solar or wind energy. Teams 1 and 3 also contemplated an electric or circular saw, but Team 1 wanted to use hydropower, and Team 3 wanted to use solar. Team 1 considered a lawnmower powered by biomass, and Team 3 wanted to power the lawnmower using solar energy. Another similarity between Teams 1 and 3 was that these teams chose to brainstorm using all the suggested renewable energy sources. This approach was different from the approach of Teams 2 and 4, where they only brainstormed ideas using the renewable energy source reported on in the research report. Teams 2 and 4 did not have as many common ideas for their respective brainstorming inventories, but they both considered the hydro-powered urinal for their final design solution. What led these teams to gravitate to certain ideas and reject others was their perception of whether or not they could make the design functional. For example, when teams considered the functionality of their designs, they would ask one another: "But how would we build it?" and "How would we make it work?"

Functional final design solutions Across all teams the common parameters that students focused on were the functional and possible/applicable parameters. All teams set goals and worked toward making a functional device and arrived at what they perceived to be a final functioning model. How students perceived the feasibility of their design solutions differed between teams. For example, Teams 2 and 4 both conceived the idea to produce the hydro-powered urinal, but Team 2 rejected this solution because they were not able to conceptualize how it would ultimately function. Team 4 on the other hand did not work under the assumption that the idea was unfeasible based on functionality and chose the hydro-powered urinal for their final design solution on the basis of the interesting/innovative parameter. Students who considered an idea a "bad-ass idea" or an "awesome idea" focused on creating an interesting or innovative design solution.

Another common theme across teams was the team's perception of the feasibility to construct a scalable model of the design idea. Teams 1, 2, and 4 built scaled models of a canopy, bridge, and urinal, while Team 3 was the only team to build their design to scale. This concept of scalability affected how teams chose or eliminated design ideas, but was not consistent in how teams scaled their designs. For example, both Teams 1 and 2 considered a solution that utilized hydro-power to produce electricity for a yacht (Team 1) or sailboat (Team 2), but rejected these ideas because they perceived them to be unfeasible when considering building the final model. Teams 1 and 2 built their final designs, the solar canopy and the hydro-powered bridge light, as scaled-down models. Team 1's perception of functionality, with respect to considering final design, is illustrated in their discussion on utilizing the turbine of the yacht during the brainstorming session:

> Eddie: That turbine that spins is going to just collect the energy from the spinning water. You're already using energy to spin it, so this is just collecting it sort of. And that collected energy, you can use it to power the electricity on board.
> Cory: I don't know how we'd do it though. It'd be pretty cool though.

End user benefit While building a working or functional model of their conceptual design was a driving force for all teams throughout design sessions, all teams considered final design ideas that met the needs of or benefited a user. The design document did not prompt

teams to consider user needs in the conceptualization stages and only prompted students to reflect on their design and the ethical implications by considering the user. The final report document prompted students to consider ethical implications by asking, “Could the user or another person be injured by your design?” Thinking about user needs was one way for students to conceptualize potential useful solutions.

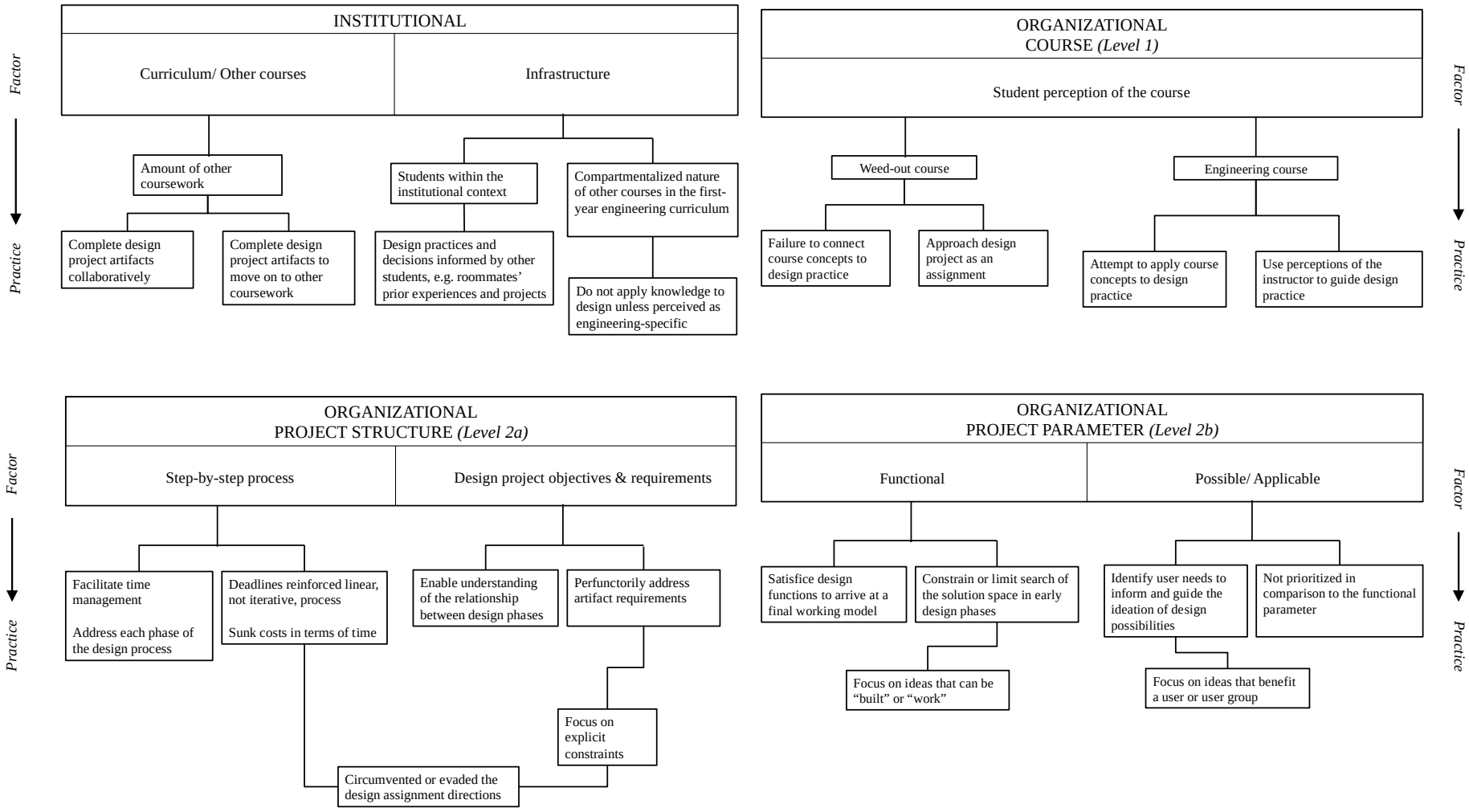
Team 1 framed their decision to design the solar canopy on the basis of one team member’s need for power outlets when lounging by the pool or beach. Similarly, during the brainstorming phase, Team 3 referred to personal experiences that included working on a deck over spring break and needing tools that did not lose power or could be easily recharged. Teams 2 and 4 commented in their interview that they chose ideas that were beneficial to certain user groups. While Team 4 felt restricted in terms of building a working model and staying under budget, they perceived their design as beneficial on a larger scale.

Post hoc actions All teams took post hoc actions regarding the other parameters not discussed until the final design report. The design document prompted students to write about the other parameters in the final report. A section in the final design report asked students, “Please discuss the ethical implications of the design you chose.” Teams did not initially identify or explain these parameters in relation to their design. The struggle to explain the ethical parameters of their design generally resulted in team members suggesting to one another that they needed to “make up” material to address these concepts in the report. Team 3’s discussion during the final design report session illustrated how students thought about the ethical considerations.

Andy: Help me out with the ethical —
Brett: Just start making stuff up.
Andy: Utilitarianism and deontology.

One outcome of prompting design teams to think about design parameters was that teams discussed the overarching issue of sustainability and environmental effects. Later in Team 3’s final design report session (after the discussion on making up material to write about ethics), they continued to think about their design in terms of its effect on the environment.

The findings from the cross-case analysis indicated that several aspects were common across teams and affected student practice in either an enabling or constraining nature. Figure 2 provides a comprehensive representation of the contextual factors and their resulting effect on design practice in first-year engineering design teams observed in this study.


Factor
Practice
INSTITUTIONAL
Curriculum/ Other courses
Infrastructure
Amount of other coursework
Complete design project artifacts collaboratively
Complete design project artifacts to move on to other coursework
Students within the institutional context
Design practices and decisions informed by other students, e.g. roommates' prior experiences and projects
Compartmentalized nature of other courses in the first-year engineering curriculum
Do not apply knowledge to design unless perceived as engineering-specific
ORGANIZATIONAL
COURSE (Level 1)
Student perception of the course
Weed-out course
Failure to connect course concepts to design practice
Approach design project as an assignment
Engineering course
Attempt to apply course concepts to design practice
Use perceptions of the instructor to guide design practice
ORGANIZATIONAL
PROJECT STRUCTURE (Level 2a)
Step-by-step process
Design project objectives & requirements
Facilitate time management
Address each phase of the design process
Deadlines reinforced linear, not iterative, process
Sunk costs in terms of time
Enable understanding of the relationship between design phases
Perfunctorily address artifact requirements
Focus on explicit constraints
Circumvented or evaded the design assignment directions
ORGANIZATIONAL
PROJECT PARAMETER (Level 2b)
Functional
Possible/ Applicable
Satisfice design functions to arrive at a final working model
Constrain or limit search of the solution space in early design phases
Focus on ideas that can be "built" or "work"
Identify user needs to inform and guide the ideation of design possibilities
Not prioritized in comparison to the functional parameter
Focus on ideas that benefit a user or user group

Discussion

Findings from this study show that institutional and organizational contexts in conjunction with aspects of the design project both constrained and enabled student design practices. From team discussions and their decision-making practices, we found that students were never fully removed from the institutional or organizational contexts and attempted to fit the design project into their other practices as students. For example, students were continually cognizant of their responsibilities regarding other coursework and utilized practices such as time management and working collaboratively to manage the responsibilities required by with their curriculum. We also found that students found it hard *not* to take a student perspective while designing. They saw the design project and associated design assignments as academic exercises. The students' perspective influenced their actions, as when they perfunctorily created design assignments. Students' attempts to meet the design project requirements also facilitated their understanding of how the phases of the design process related to one another.

The findings from this study demonstrate the importance of taking into account the educational setting when studying engineering design. Experimental studies on design education are limited by the context in which they take place, a context that often does not account for factors external to the design process. If engineering education researchers and practitioners intend to translate research into practice, they must consider the effect of aspects related to the larger, overarching contexts. Since our goal here is primarily theory explication and development through valid and reliable analysis, we have provided a generalizable explanation for what we observed. Educators can use similar situations to understand the contexts through the lens of nested structuration, and learn lessons about interpreting similar contexts and activities. For instance, in his study of engineering researchers working in a research and development laboratory, Johri (2015) similarly found that researchers' impressions of others' expertise were based both on micro-level situational interactions and the larger organizational-level context in which they worked. Overall, this study fits in a long line of literature in organizational theory that calls specifically for understanding social behavior through examination at both the micro (individual) and macro (organizational) levels (Bamberger, 2008; Klein, Tosi & Cannella, 1999). Given the interpretive nature of this study, alternative explanations for the findings are possible, but the possibility does not reduce the validity of our analysis or the usefulness of our findings, especially the relevance of nested structuration for understanding student behavior within educational settings. Irrespective of the specific theory used to interpret the our findings, what is crucial to understand is that the behavior of students, as well as the interpretive explanations of student behavior, is situated in their specific contexts (Johri & Olds, 2011). For instance, from a social psychology perspective, the findings from this study can be explained in terms of students' self-directed autonomy: their ability to exert influence and control over the design definition and process (Kusano & Johri, 2014). It can be argued that formal experiences, rather than informal experiences, limit students' autonomy and that the institutional and organizational constraints act as inhibitors of learning since they reduce the students' ability to experiment.

Our research should not be seen as overly critical of how students approach design or how the contexts within higher education are structured, but as a realistic view of student engineering design practice that can be used to enhance both teaching and research. Although the teams' final designs did not represent significant ingenuity or creativity, their discourse showed significant creative strategy in how they accomplished the body of coursework required of a first-year engineering student. Students' ability to adopt creative strategies to work on a design

project is promising in terms of preparation for their future careers and even their remaining educational career. The student strategies or approaches that we observed in this study align with Dannels's (2000) and Paretti's (2008) findings on student practice and transfer of adaptability to the professional context. Students, designers, or engineers will always be working within the contexts of an institution and an organization, and they will be required to complete many tasks. While students may not take the instructor's view on valuing design practices as they relate to good design or best practices, they can evaluate their time available, identify what is required or expected of them, and determine how to optimize the available resources. By working under a set of given constraints, they also accomplish these tasks collaboratively as a design team.

In spite of neat compartmentalization of courses and projects, where university courses and projects typically do not have much overlap or integration, students are not cognitively disconnected from their other experiences and other students' experiences. For students, the higher educational setting is a seamless environment where they undertake multiple curricular and co-curricular activities. The students' actions raise concern that students will be unable to integrate their many educational activities and experiences. For example, students learn technical concepts in other courses, but never apply them to their designs, and typically fail to do so because applying content from other courses is not required by the project or structured that way. The structures that make up the contexts allow for multiple activities to take place, but do not force students to integrate those activities. As design educators work towards improving design education (Borgford-Parnell, Deibel, & Atman, 2010), they need to pay more attention to how the context of students shapes their design practices. Situating the activities within the context will also help to align school and workplace design practices (Atman et al 2007).

Suggestions for Design Educators

While design educators have limited control over the larger context in which they work, this study suggests that they can use the features of the higher educational contexts and findings from this study to more effectively enable students to produce better designs by facilitating a better design process. We identify four main impediments to students' achieving the desired learning objectives for design activities and on the basis of our results, offer solutions to facilitate and enhance design practice. Table 5 lists each of the impediments and summarized solution.

Table 5 Suggestions for Design Educators

Impediment	Solutions
Familiarity with the institutional context	Incorporate the STEM integration approach and cohesion of central concepts
Linear approach to design	Promote iteration of design phases through the structure of the project
Cognitive inertia and sunk-costs related to time	Promote easier transitioning between design phases
Focus on explicit constraints	Facilitate conceptual understanding through design assignments requirement definition

Impediments and Solutions

Limitations of the institutional context The observed first-year students were familiar with the institutional context; that is, they were accustomed to the practices relating to taking courses, managing time, and coursework, but were not familiar with the design context. Design educators can use students' interaction with the institutional contextual factors by helping to reinforce or integrate material from other courses as part of design, rather than treating the design project as a separate entity. One program designed to promote the connections between science, technology, engineering, and mathematics concepts is STEM integration. The STEM integration approach is based on producing and maintaining the cohesion of STEM concepts through representations, objects, activities, and social structures (Nathan et al., 2013). This approach is applicable to the institutional context where first-year engineering students typically take science-, technology-, and mathematics-related courses at the same time as they learn design.

Linear approach to design Design teams in this course liked the step-by-step, or gradual, nature of the design project because the deadlines facilitated time management within the context of their other work. This feature also prompted students to address each phase of the design process by completing assignments associated with a phase, such as researching, brainstorming, and prototyping. At the same time, students approached the design assignments as isolated academic exercises with deadlines, rather than integrated phases of the design process. The project's deadlines promoted a linear approach through the process and did not encourage students to revisit phases. Educators structuring a project should consider how to facilitate students in iterative design and allow for flexibility within and across design phases (Kramer, Daly, Yilmaz, & Seifert, 2014; Liu & Lu, 2014). The benefits of frequently transitioning between design steps and being able to spend an adequate amount of time on the product realization and final stages of design are apparent in Atman et al.'s (1999) comparative study of freshmen and senior designers. They found that students who transitioned frequently between design steps throughout the process produced higher quality designs.

Cognitive inertia and sunk costs related to time Students in this study did not have opportunities to repeat design phases because of the rigid project deadlines for each design assignment. Infrequent opportunities to repeat phases or steps led to cognitive inertia (reluctance to revise their assumptions) and sunk costs (unrecoverable costs) in terms of time. In other words, once students made design decisions and submitted their assignments, the project timeline did not allow for the reversal or redesign ideas because assignments or materials associated with the next phase of the design process were typically due the following week. Students had little control over the project deadlines; a lack of control over project submission deadlines led the students to work strictly under the time constraints. Many decisions could not be reversed due to time constraints, so students often tried to evade meeting the requirements, and fabricated or improvised various sections of the final design report. To differentiate the impediments related to the linear approach to design and impediments related to cognitive inertia and sunk costs, we suggest that the solution of restructuring deadlines may not always assist students in overcoming cognitive inertia or sunk costs. For example, students who commit to design ideas or approaches early in the design phases and avoid rethinking fundamental ideas regarding their design could perfunctorily create or revise design assignments in order to meet the requirements of the project. Design educators can change the structure of the project in order to promote a more iterative and contextualized approach to design. Students who are encouraged to revisit design phases will have the opportunity to redesign and make changes, as well as engage in opportunities for learning engineering (Adams, Turns, & Atman, 2003b). Students who revisit design phases and

design assignments can have a better understanding of the purpose of each design phase. The implications of revisiting or allowing for more repetition of design activities are supported by Newstetter's (1998) study on student design teams' views on the discrepancies between the "learning view" of the instructor and students' "institutional view" (see Literature Review section on constraints). Students' conceptual approach to design was based on their view and perceptions of the institutional context, and students framed design activities as another set of methods or procedures, or procedures to be mastered or faked. In order to give meaning to design activities, Newstetter recommended repeating activities and placing them in context outside of the classroom and institutional view.

Focus on explicit constraints The data in this study showed that students often focused on explicit constraints given in the project document when they did not understand the design assignment requirements. For example, design teams often had trouble interpreting or defining the design problem, so they leaned toward explicit constraints such as the budget. Similar findings from Newstetter's (1998) study identified failed affordances, or failed action possibilities, as a result of students' approach to design. Newstetter (1998) also found that students viewed design activities as tasks to be completed rather than opportunities for learning about design. Students still met or attempted to meet the given requirements of the design project and fit the given constraints. Our findings suggest that first-year students will work within the explicit statements or requirements of the design document. For example, if students were uncertain about the requirements and purpose of the prototype, they often based their practices related to constructing the prototype on previous students' experiences. Design educators can capitalize on this finding by designing requirements that better fit the intentions or objectives of the design assignments and the design process. For example, educators should be more explicit when communicating the objectives of prototyping as part of the design process.

The research presented in this article demonstrates the influence of the organizational and institutional contexts on student practice. Engineering educators should carefully reconsider the potential effect of their design pedagogy – specifically, design projects – when they are implemented within a higher education context. Educators can also incorporate methods to overcome those impediments to student learning in design that are caused by the institutional and organizational contexts.

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Appendix

Interview Protocol

This appendix is provided to allow readers to view the format and questions asked in the team interviews.

Form:

Semi-structured; posing questions and appropriate follow-up questions

Begin:

Explain to students the interview will be regarding their experiences with the design project as well as the course and that all discussions will be confidential. Have students begin by writing down their thoughts and experiences with both the design project and the course.

Interviewer:

What are some of the things that come to mind? What stands out about the project?

[Provide time for students to write topics down on paper individually]

1. Design Project

a. What role did each of you play?
d. What did you think about your final project?
e. How did you decide on your final solution?
f. What were your other options and how did you choose?
g. What did you think about the other projects in the class?
i. What would you change about your project or the structure of the project in general?

2. Technology Use (Data collected regarding technology use was not used or analysed in this study.)

a. How did you work on the project?
b. Did you use technology?
c. Did you use the tablet feature of the PC?
d. How did you use the tablet?
e. How did you sketch?
g. What did you think of [[course instructional software]]?

3. Overall

a. What do you think design is?
b. What do you think engineering design is specifically?
c. What are your intended majors? Do you think that made a difference?
d. Did you encounter any conflict during this experience? Describe the conflicts that arose.

4. Closing

a. What did you learn from this experience?
b. Is there anything else we did not cover you wanted to discuss?